Multiplying polynomials

1

a

i y^2 ii $3y$ iii $7y$ iv 21

b $y^2 + 10y + 21$

2

a $ab - 6a + 9b - 54$

b $28rs + 8r + 49s + 14$

c $x^2 + 7x + 10$

d $x^2 + 12x + 35$

e $v^2 + 18v + 80$

f $x^2 + 3x - 10$

g $x^2 - 5x + 6$

h $m^2 - 3m - 18$

i $5m^2 - 7m - 24$

j $35w^2 + 39w + 10$

k $42y^2 + 6y - 36$

l $4c^2 - 56.2c + 2.8$

m $\frac{1}{9}y^2 + \frac{1}{36}y - \frac{1}{24}$

n $-90y^2 + 26y + 48$

o $9y^2 + 90y + 144$

p $7x^2 - 42x - 189$

q $-x^2 - 7x - 10$

r $-5y^2 - 45y - 100$

s $-4y^2 + 12y + 112$

t $-75x^2 + 110x - 40$

u $6x^2 - 12x - 18$

v $6x^{10} + 13x^5y^4 + 5y^8$

w $-30x^2 - 42x + 34$

x $-8x^2 + 40x + 72$

y $18x^2 - \frac{69x}{2} - 8$

z $-4x^2 - 6x + 32$

3 $\left(12x^2 + 144x + 384\right)\ \text{m}^3$

4 $42x^2 + 59xy + 20y^2$

5

a $(h + 9)\ \text{cm}$

b $(w + 8)\ \text{cm}$

c $\left(4h + \frac{9w}{2} + 36\right)\ \text{cm}^2$

6 $\left(128y^2 + 48y\right)\ \text{cm}^3$

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Additional questions

19

a 4 and -28

b 1 and -11

c Answers may vary. 2, 6, and 18 is one possibility. 3, 4, and 18 is another.

20

a The new dimensions are $(5+2x)$ and $(7+2x)$

b $4x^2+24x+35$

c $4x+24$

21

a The new dimensions are $(12-2x)$, $(12-2x)$, and $(5.5-2x)$

b Jerome will have a volume of $-4x^3+59x^2-276x+396$ left to fill.

22

a $(10x)(10x)=100x^2$, but Anette put $100x$ instead. Her final answer should have been $100x^2+20x+30x+6=100x^2+50x+6$

b

	x^2	$10x$	2
$10x$	$10x^3$	$100x^2$	$20x$
3	$3x^2$	$30x$	6

c $10x^3+103x^2+50x+6$

23 The result will be a polynomial with degree 7.

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24

a Agree. The maximum number of terms that can be produced by multiplying two binomials is $2 \times 2 = 4$ if no terms combine.

b Answers will vary. A monomial multipled by a five term polynomial will always results in a five term polynomial. Another possible soluton is a binomial multipled by a trinomial that results in a $4th$ degree polynomial with exactly one pair of terms combining.
For example, $(x^2 + 3)(x^2 - 2x + 1) = x^4 - 2x^3 + 2x^2 - 6x + 3$

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